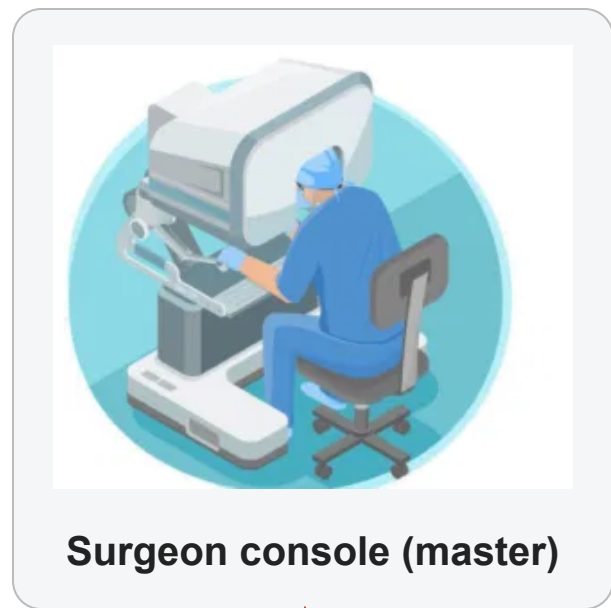


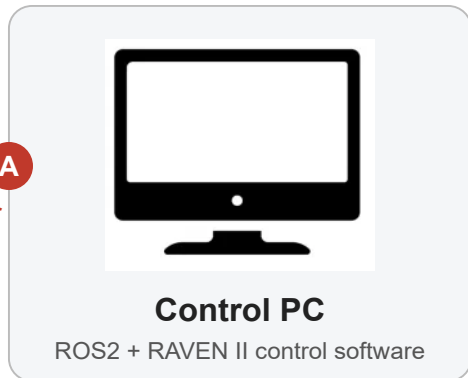


Teleoperated operation
surgeon sends commands
via console

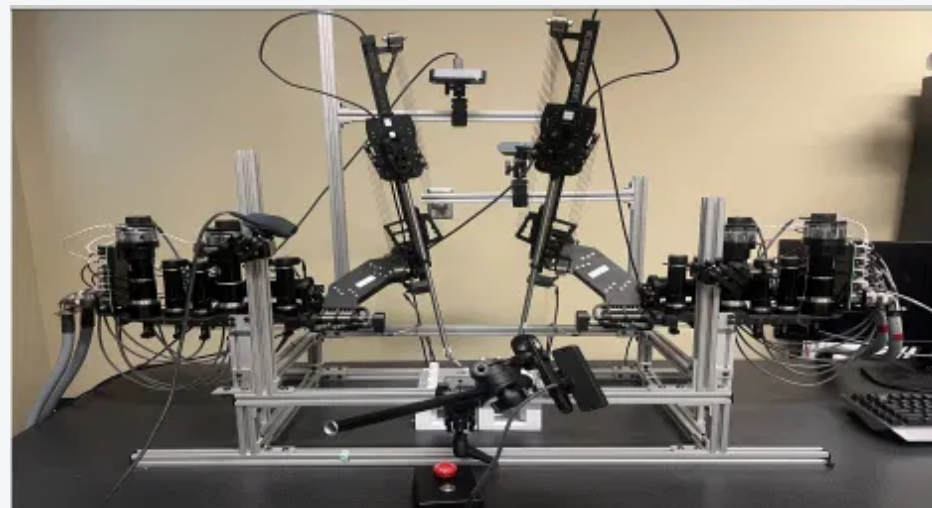
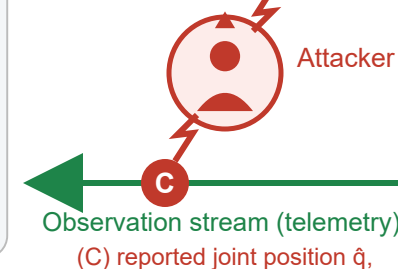
Autonomous operation: PC-generated trajectory

Master-tool pose / pedal state (UDP)

(A) desired pose q_d , written after
the command is received



Command stream
(B) motor torque τ , before DAC write



RAVEN II (slave)

2 arms x 7 DoF, motor encoders, motor drivers, USB I/O boards

Attack surface and detection target of this work

Patient / phantom side: instruments, ports

Deploy the detection on this host

D Telemetry monitor (detection point)

passive tap on the telemetry stream; no change to the control software

Window-scale path (227 ms)

Torque-to-motion residual (Eq. 5-6): reported motion vs. reported torque, sees (C)
Servo tracking error $q_d - \hat{q}$: motion vs. command, sees (A), (B)
Learned detector (two-layer LSTM, Eq. 3): scores each window

Session-scale path (hourly, cumulative drift)

Paired-motor cable-coupling consistency $e_1 = a d + b$ (Eq. 4),
against an external encoder reference

Alarm when a residual exceeds its threshold, set by the alarm budget (two per hour)

Injection patterns (per joint)

Step



Ramp



Noise



added to the true signal $x(t)$: $x(t) + d(t)$

Alarm + attack class +
affected joint

shown on console / e-stop request

